Factors, Multiples and Patterns — Diagnosing a Rule That Was Guessed From Too Few Terms

Grades 4–5

A rule that fits the first two or three numbers is only a **guess**. A guess that is then used everywhere is an **overgeneralized rule**.

- In each problem, someone states a rule and the evidence they checked.
- Test the rule on a term they did *not* check, or find one **counterexample** — a single case where the rule fails.
- Then answer the question with the *repaired* rule. Show your testing.

1. Pattern A: 3, 6, 9, 12, ...
Kip checks only the first two terms and writes the rule "*double the term before*" — and 3 doubled really is 6.
Test Kip's rule on the third term. Does doubling 6 give the third term of Pattern A? Then use the pattern's real rule to find the 7th term.

Answer: __________________

2. Pattern B: 1, 4, 9, 16, ...
Dee writes the rule "*add* 3" because $1 + 3 = 4$.
Show that Dee's rule fails at the third term. The real pattern is the square numbers (1×1, 2×2, 3×3, ...). Find the 7th term.

Answer: __________________

3. Find and fix. Nia claims: "*Every multiple of* 4 *is also a multiple of* 8." Her evidence is 8, 16 and 24 — all multiples of 8.
Ravi answers that 12 is a counterexample. Divide 12 by 8 and write the **remainder**. A remainder that is not zero proves Nia's rule is overgeneralized.

Answer: ________________

4. Find and fix. Pattern C: 5, 10, 15, 20, ...
Tam writes: "*The terms are the multiples of* 5*, so every term ends in* 5." The first half of Tam's rule is right; the ending-in-5 part is overgeneralized from the first term only.
Cross out the wrong half of Tam's rule. Then use the correct half to find the 12th term.

Answer: ________________

5. Pattern D starts at 4 and adds 4 each time. Bea says the rule is "*double the term before*" — which gives the same first two terms.
Work out the 6th term of Pattern D, and the 6th term Bea's doubling rule would give. Then write $<$, $>$ or $=$ between them:

Pattern D's 6th term ________ Bea's 6th term

6. Find and fix. Ollie claims: "*Every odd number is prime.*" His evidence is 3, 5 and 7. Find the smallest odd number greater than 7 that is *not* prime, and prove it is not prime by writing it as a product of two smaller whole numbers, both greater than 1. Write the product's value on the answer line.

Answer: ________________

7. Pattern E: 1, 2, 4, 7, 11, ...
Ines writes the rule "*double the term before*" because 1 doubles to 2 and 2 doubles to 4. Show the exact term where Ines's rule first breaks. The real pattern adds 1, then 2, then 3, and so on — one more each step. Find the 8th term.

Answer: ________________

8. **Explain.** Rosa claims: "*Every number that ends in* 0 *is a multiple of* 20." Her evidence is 20, 40 and 60.

(a) Explain, in your own words, why three examples that work do not prove Rosa's rule.

(b) Give one counterexample to Rosa's rule and show why it is a counterexample.

(c) Write one sentence a classmate could use as a habit for testing any new pattern rule.